PAPER ID-311034

Subject Code: BME071

Roll No: | | | | | | | | | | | | | |

BTECH
(SEM VII) THEORY EXAMINATION 2025-26
ADDITIVE MANUFACTURING

TIME: 3 HRS **M.MARKS: 70**

Note: Attempt all Sections. In case of any missing data; choose suitably.

SECTION A

1. Attempt *all* questions in brief. **02 x 7 = 14**

Q no.	Question	CO	Level
a.	Define additive manufacturing and mention any two advantages over conventional manufacturing processes.	CO-1	K1
b.	State the distinction between additive manufacturing and computer numerical control machining.	CO-2	K1
c.	Write any two material handling issues encountered in additive manufacturing systems.	CO-3	K1
d.	What is meant by scan pattern in vat photo polymerization process?	CO-4	K1
e.	State any two quality aspects that influence material selection for additive manufacturing..	CO-5	K1
f.	State the importance of intellectual property in additive manufacturing.	CO-1	K1
g.	Write any two applications of additive manufacturing in the medical field.	CO-2	K1

SECTION B

2. Attempt any *three* of the following: **07 x 3 = 21**

a.	Explain the distinction between direct and indirect additive manufacturing processes with suitable examples.	CO-1	K3
b.	Describe the role of computer aided design technology in the development of additive manufacturing systems.	CO-2	K3
c.	Explain the working principle and process parameters of powder bed fusion process.	CO-3	K3
d.	Discuss additive manufacturing design strategies and their impact on product quality.	CO-4	K3
e.	Explain the scope of additive manufacturing in aerospace and automotive applications.	CO-5	K3

SECTION C

3. Attempt any *one* part of the following: **07 x 1 = 07**

a.	Explain the types of additive manufacturing technologies and their applications in prototyping and tooling.	CO-1	K3
b.	b. Analyze the advantages and limitations of layer manufacturing processes such as polymerization and extrusion.	CO-1	K3

4. Attempt any *one* part of the following: **07 x 1 = 07**

a.	Describe the eight steps involved in the generalized additive manufacturing process chain.	CO-2	K3
b.	Evaluate the design considerations and material handling issues in metal based additive manufacturing systems.	CO-2	K3

5. Attempt any *one* part of the following: **07 x 1 = 07**

a.	Explain vat photo polymerization process with reference to materials, reaction rates and process modeling.	CO-3	K3
b.	Discuss extrusion based additive manufacturing systems including plotting and path control mechanisms.	CO-3	K3

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6. Attempt any *one* part of the following: 07 x 1 = 07

a.	Explain software issues in additive manufacturing with special reference to the standard tessellation language file.	CO-4	K3
b.	Analyze engineering design rules and material quality aspects for additive manufacturing.	CO-4	K3

7. Attempt any *one* part of the following: 07 x 1 = 07

a.	Discuss machines used in additive manufacturing and secondary rapid prototyping processes.	CO-5	K3
b.	Evaluate future trends and business opportunities in additive manufacturing with suitable examples.	CO-5	K3

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